A Scholarly Comparison of AI Strategies in a 2D Fighting Game

Course: Artificial Intelligence Fundamentals

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The game

Aim of the project

Building multiple agents, based on methodologi es studying.

They should be able to defeat at least a human novel.

PEAS Framework

Performance

Environment

Actuators

Sensors

Performance

Outcome-based metrics

Victory rate = Games won / Total games

Environment

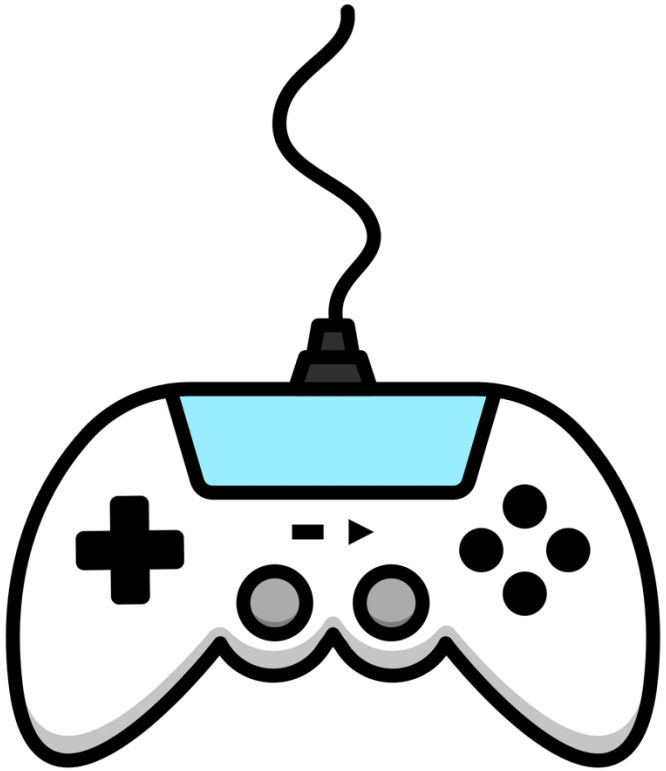
Partially observable: due to delay
Partially deterministic: some actions have certain outcomes, while others are be conditioned by random factors (Non sono certo)
Discrete: actions are finite and time is binned of length 15 frames

Environment

Episodic: 3 rounds, each lasting 60 seconds

Bi-agent: since is 1vs1 game

Stationary: the environment's dynamics do not change over time



Actuators

Movements

Dash

Basic attack

Combo attack

Defense

Dodge

Sensors

2D coordinates of the agent on the map

Life points

Energy

Oppo status

Round time

ProBLog[1] Agent

Introduction: Probabilistic logic programs are logic programs in which some of the facts are annotated with probabilities.

Its knowledge base can be represented as Prolog/Datalog facts

Enriched with probabilities:
0.4::asthma(X) :- smokes(X).

Which means, if *X* is a smoker then with 40% is asthmatic

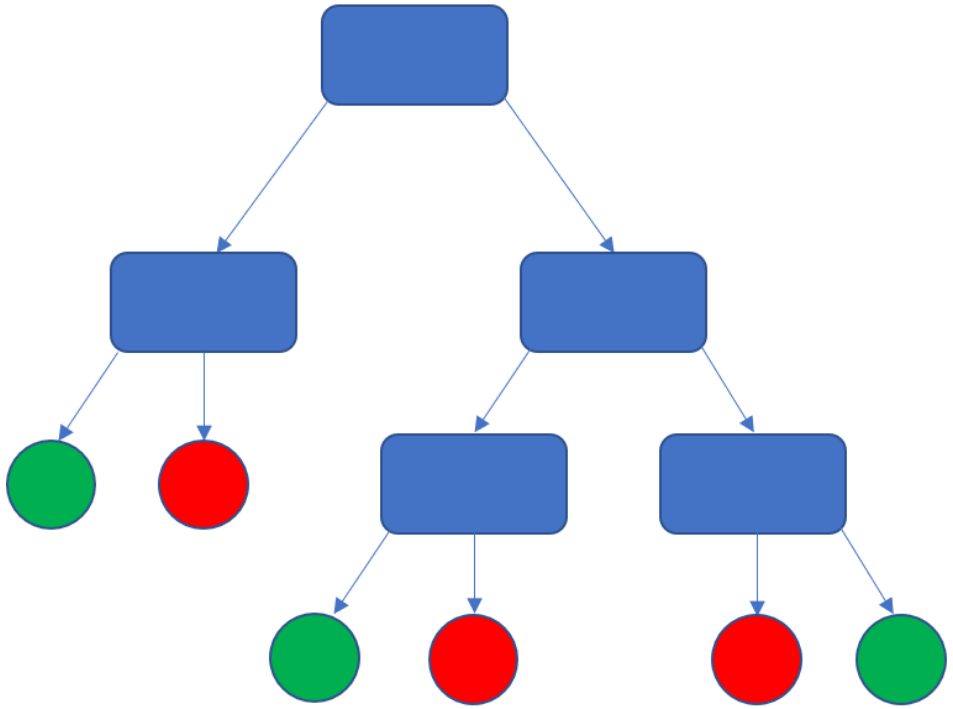
1. https://dtai.cs.kuleuven.be/problog/index.html

ProBLog Agent

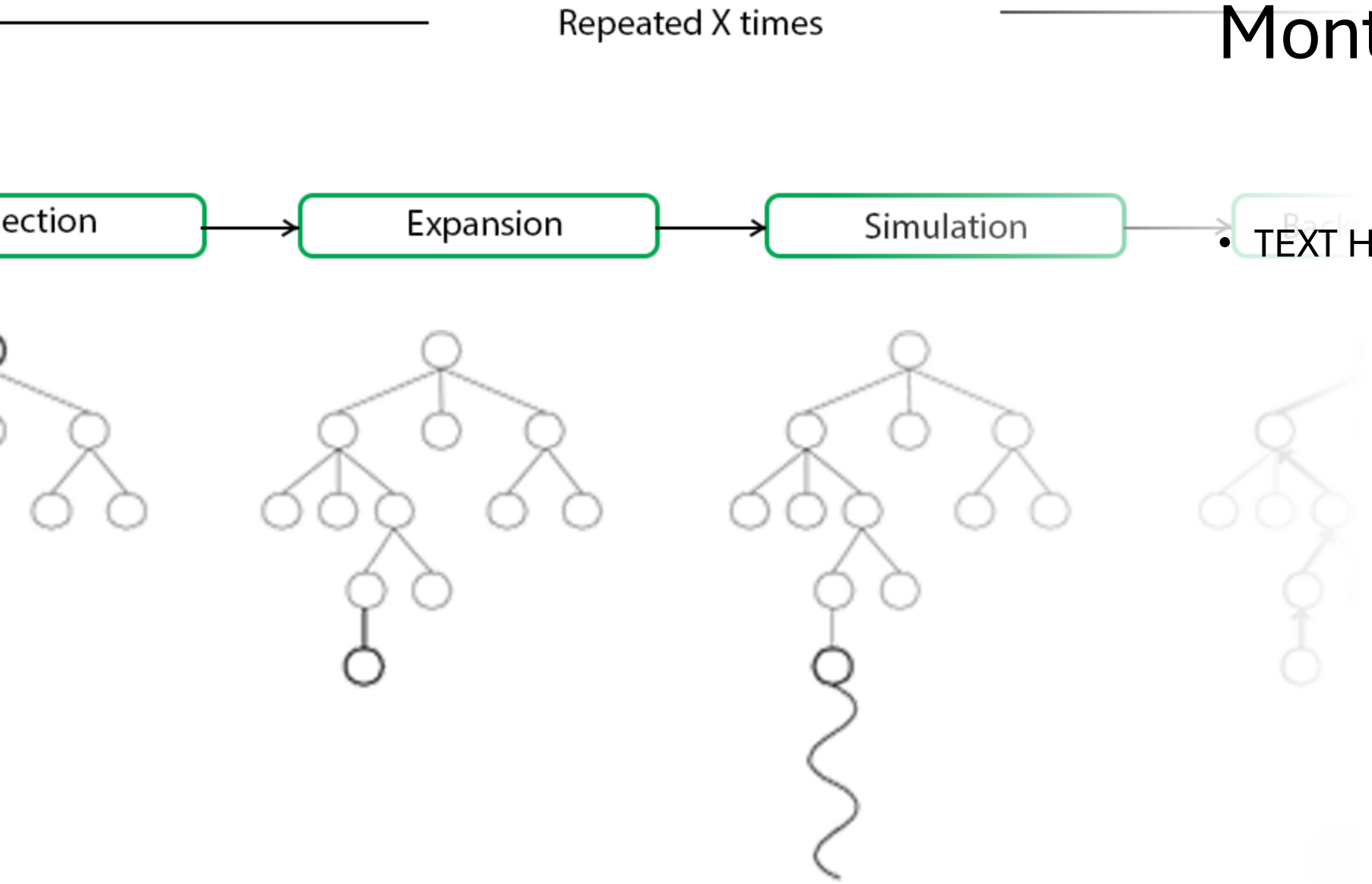
KB can have unknown probabilities of the form:
`t(_)::p_alarm1.`

This indicates that the probability of this fact is to be learned from data.

For more information, visit: https://problog.readthedocs.io/en/latest/index.html.



Montecarlo agent



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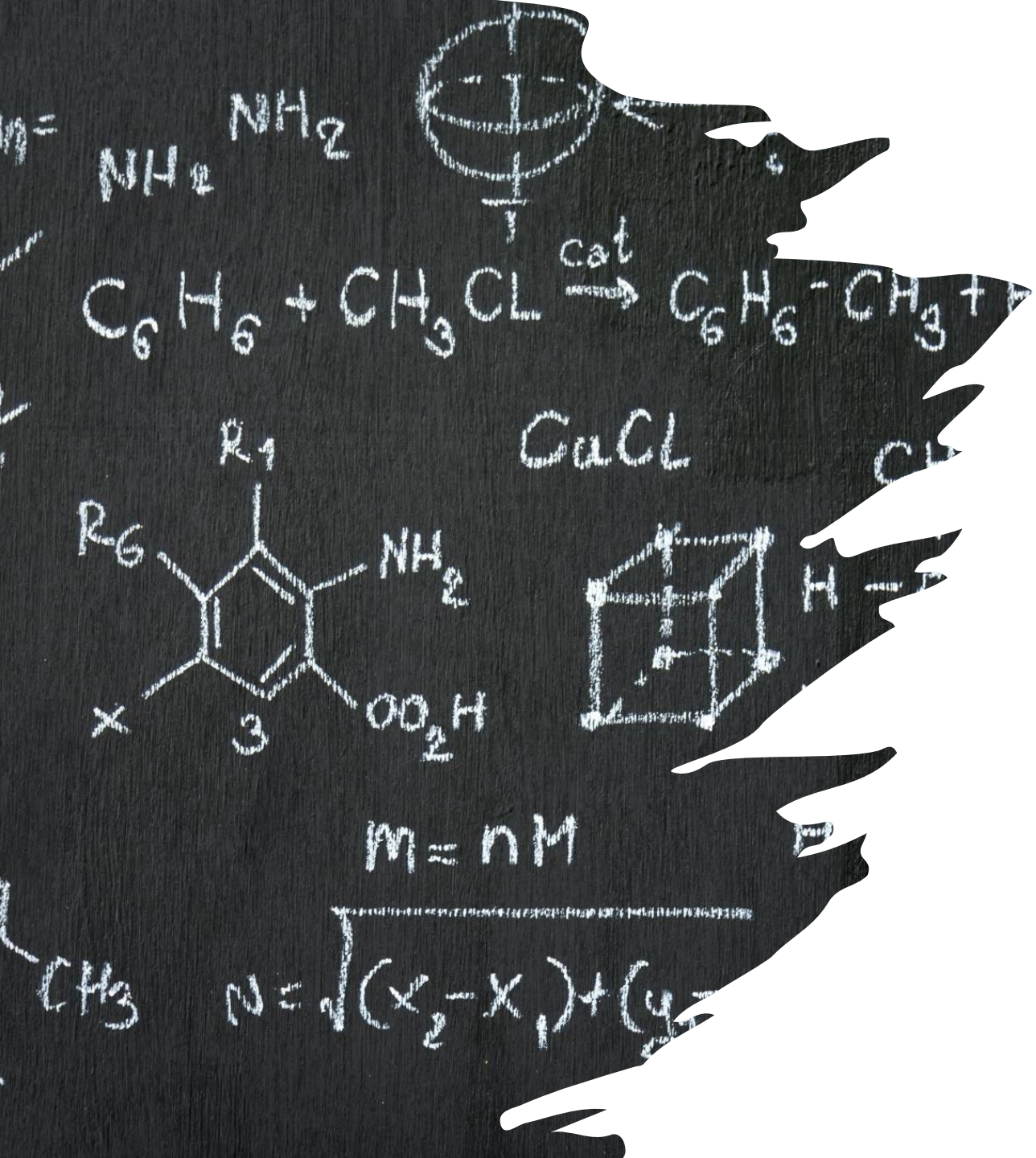
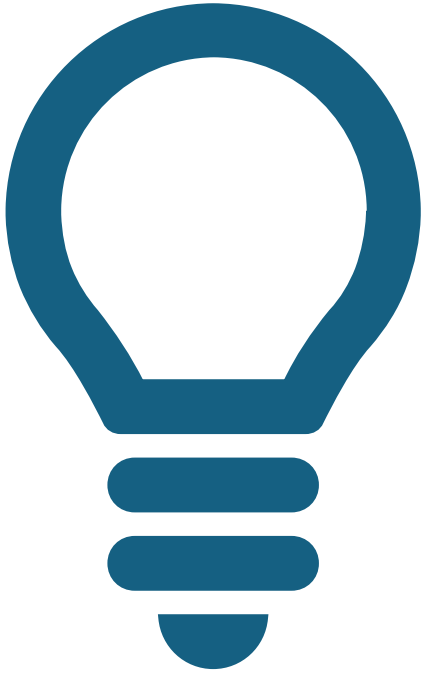


Knowledge Base

- Encoding each possible state (*stand, crouch, air, down*) with unknown probabilities, e.g: `t(_)::state(Agent, stand) :- agent(Agent).`
- Divide actions (with actual probability, sum to 1) in:
  - *High damage, Basic attacks,* e.g.: `0.28::s_action(stand_d_df_fc)`
  - *Movements* and *defense reactions,* e.g:. `0.2::m_action(back_jump).`
  - In total 19 actions out of 56 possible.
- The damage each action can inflict, and energy cost/gain, e.g: `energy_cost(Agent, stand_d_df_fc, 150) :- agent(Agent).`
- Also, the distance required to hit the opponent, e.g: `max_distance_to_hit(Agent, stand_d_df_fc, _) :- agent(Agent).`

Challenges

- 15 ms response time: We'll try exact inference: with SDD (Sentential Decision Diagrams) and with and heuristic KBestFormula, which is a ProbLog component that finds the k-best (most probable) explanations for queries
- Creating a ground, efficient and optimal Knowledge Base.
- Montecarlo complexity in time (??????)



Solutions

- Solution 1
- Solution 2
- Solution 3

Conclusions

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THANKS TO ALL

ANSWERS